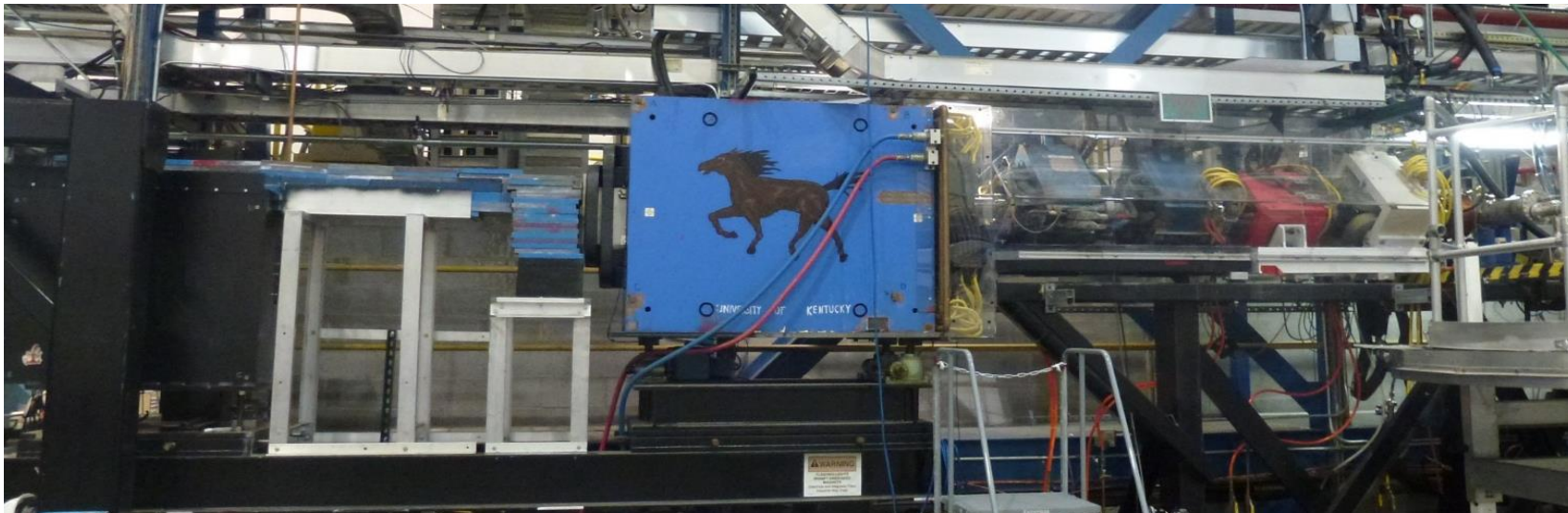


MOLLER POLARIMETER for PREX/CREX



S. Glamazdin, R. Pomatsalyuk

Systematical Error Budget for DVCS

Variable	Brute force
<i>Target polarization</i>	0.5%
<i>GEANT</i>	0.5%
<i>Target temperature</i>	0.2%
<i>Dead time</i>	0.05%
<i>Background</i>	0.3%
<i>High beam current</i>	0.2%
<i>Stability</i>	0.5%
<i>Total:</i>	1.0%

15 Moller measurement in 2014-2016 for DVCS.
Most of measurements with 8+GeV beam energy

PREX / CREX CONDITIONS

2018

Beam energy: 1 - 2GeV

Required accuracy: $\Delta P/P < 1\%$

Time to run experiment:

Relatively short (a few Moller measurements at given beam energy)

Moller polarimeter operation experience:

1GeV – no experience after upgrade

2GeV – short experience in 2015

(two measurements with straight and bad quality beam)

New target has to be commissioned. Measurements in test Lab and Kerr-apparatus can help a lot. (More details in Don's talk yesterday)

TARGET SYSTEMATICS STUDY

New target from TEMPLE University:

Has to be done in the Test lab with magnetic field:

- Magnet field mapping
- Target loader survey
- Target rotation angular resolution
- Target linear motion accuracy

Can be done in the Test lab (off line) with Kerr-effect apparatus:

- Target saturation vs field for all foils
- Target saturation vs target angle for all foils (find 90° between foil and field)
- Warping effects: target saturation vs field in different points
- Foil thickness fluctuation effects (foils thickness $<10\mu\text{m}\pm 25\%$, foils thickness $10\div 50\mu\text{m}\pm 10\%$, pinholes)

Has to be done with beam at PREX /CREX

- Target saturation test for one of the target foils
- Polarization vs Target heating / beam current

SUPERCONDUCTING MAGNET

Magnet alignment is critically important at 1 - 2GeV

Alignment test is ~6 hours of beam time

We were not successful at PREX-I

Strong influence on beam steering and Moller acceptance

Prefer 3T to minimize influence of the field

Perfect magnet alignment is needed to ramp magnet up in parallel with Moller settings

MOLLER SPECTROMETER + DETECTOR

Known alignment of quads

No data for 1GeV

Limited data for 2 GeV

Strong dependence of detector alignment

DAQ's

Helicity frequency: $120 \div 250$ Hz

- Not a problem for both FADC and old DAQ
- Hopefully, will be able to measure Levchuk-effect with FADC data triggers

New FADC DAQ has to be commissioned

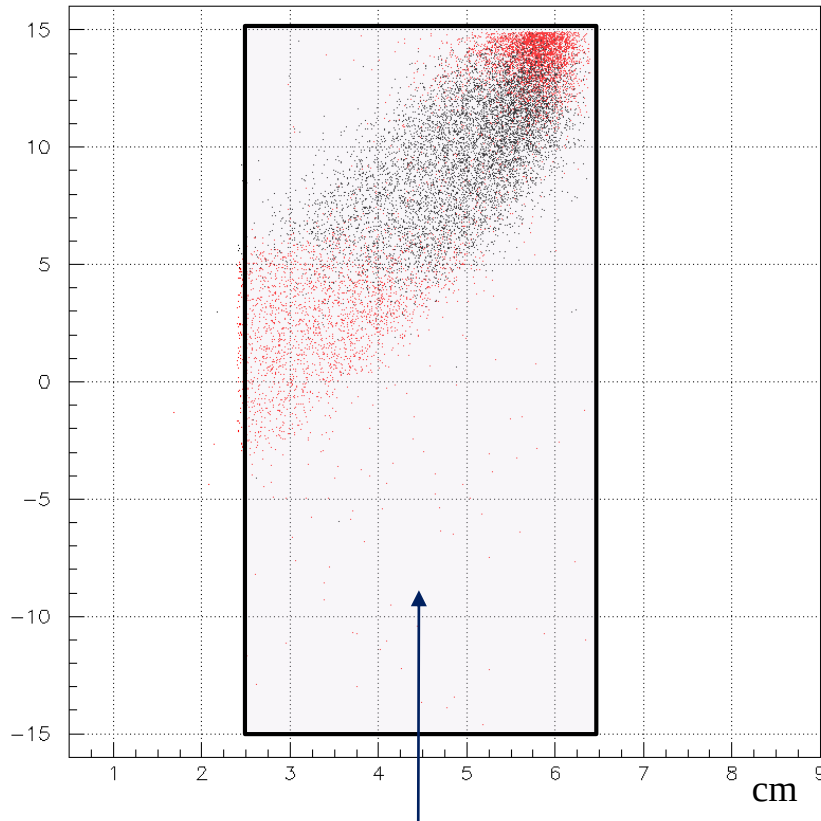
RATES

Beam energy, GeV	θ , deg	Φ , deg	Analyzing power	Single arm a.u.	Coincidence a.u.
1.0 PREX	5.5	2.3	0.7667	13	7
2.0 CREX	10	3.8	0.7418	37	23
11.0 DVCS	4	14	0.7724	74	32

Rates difference can be compensated by target thickness ($12\mu\text{m} \rightarrow 20\mu\text{m} \rightarrow 30\mu\text{m}$)

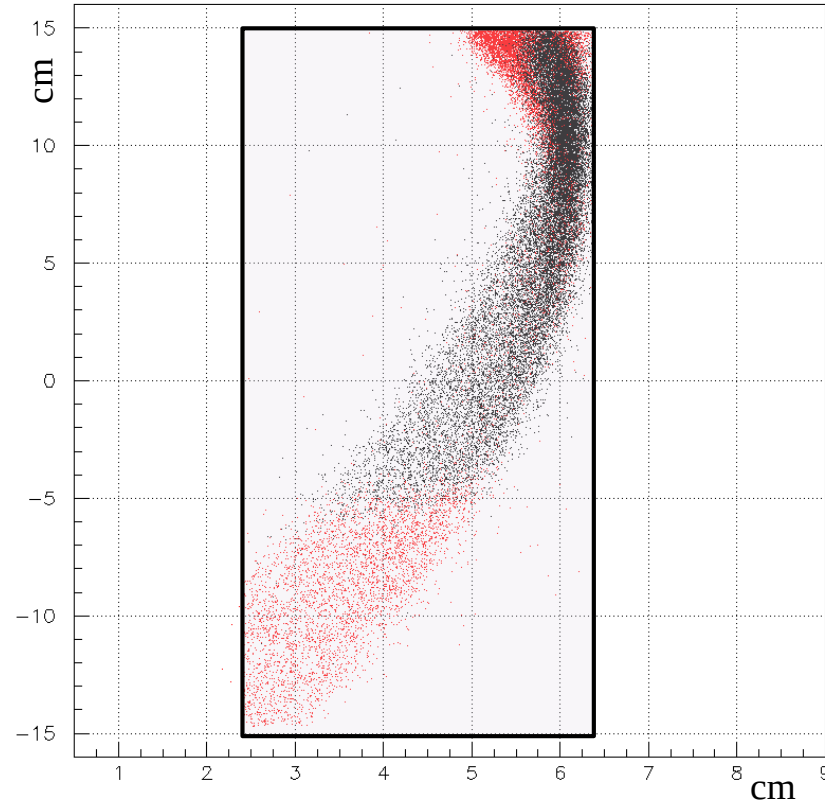
GEANT SIMULATION

1GeV

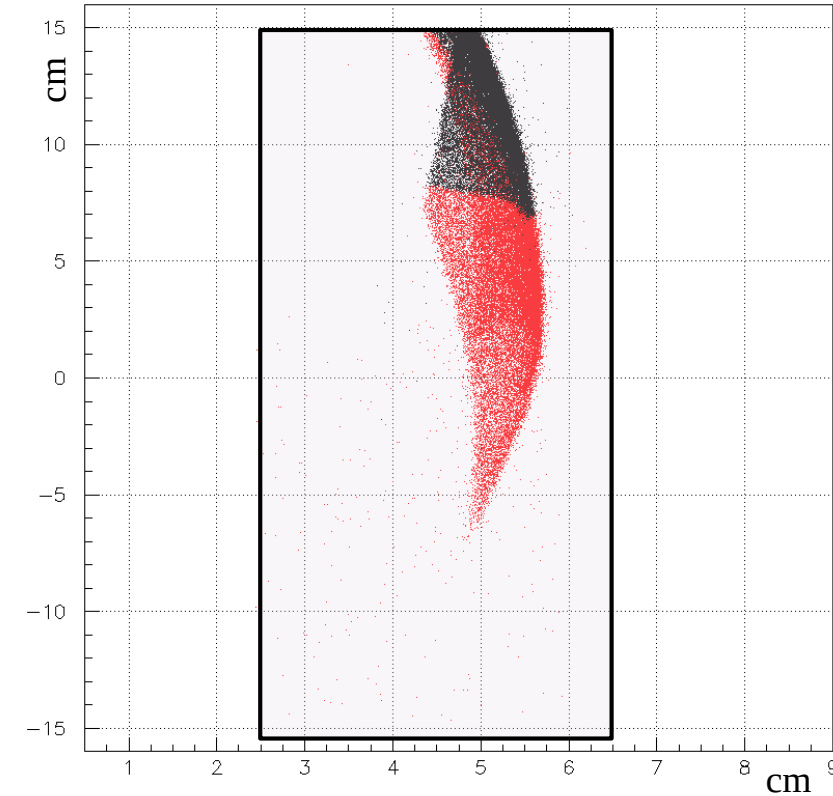


Aperture detector

2GeV



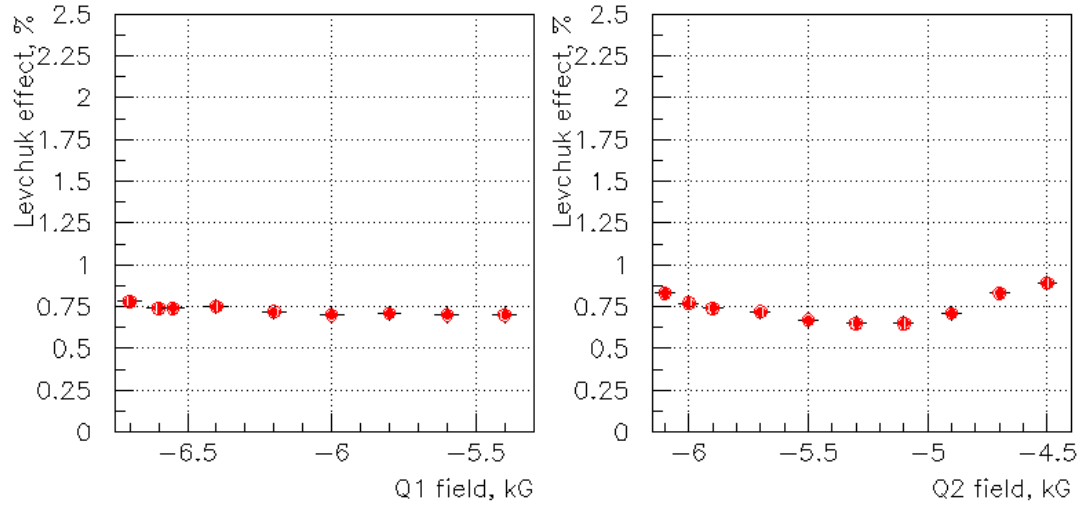
11GeV



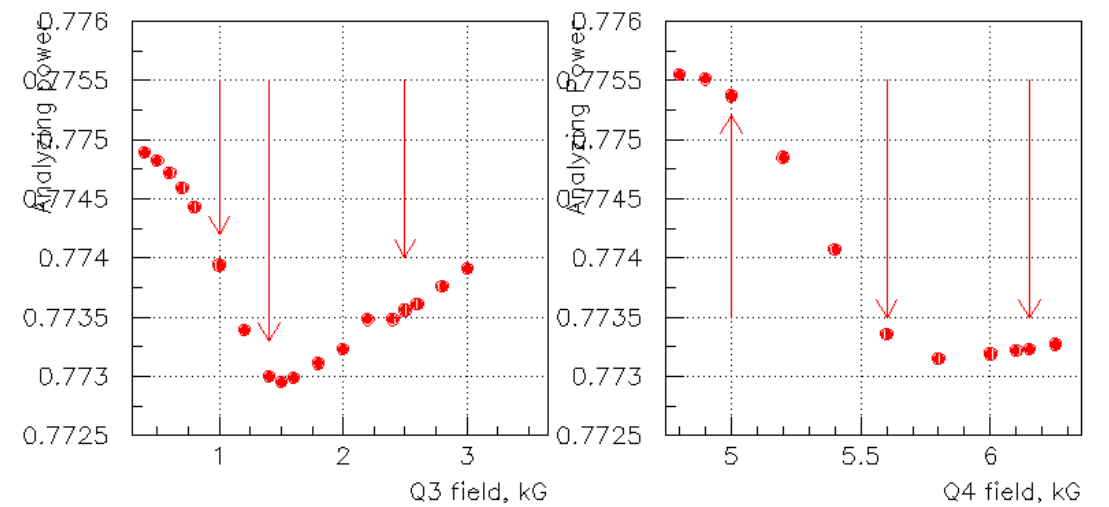
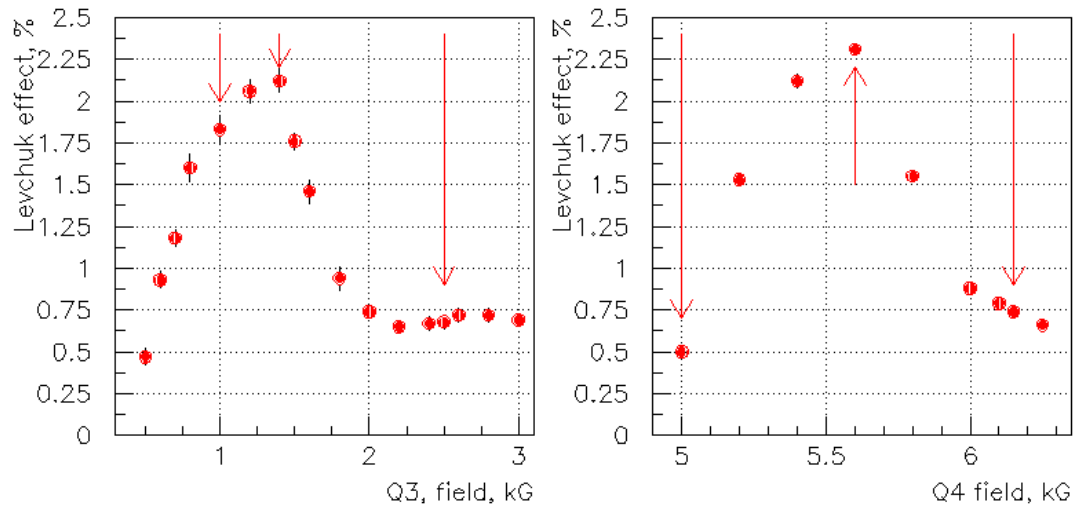
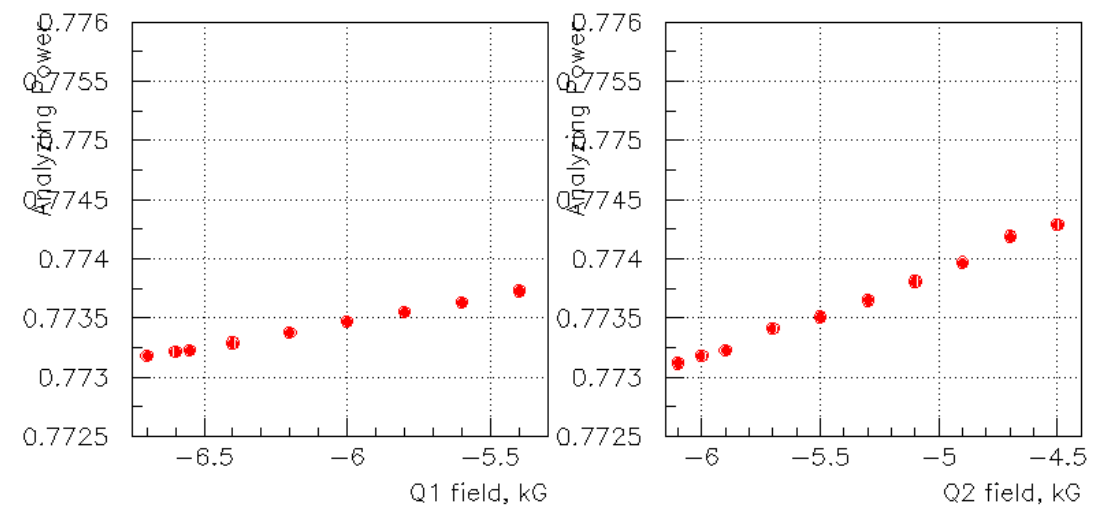
Moller events (black – single arm, red – coincidence) vs beam energy

GEANT TEST

Levchuk-effect at Ebeam=11GeV



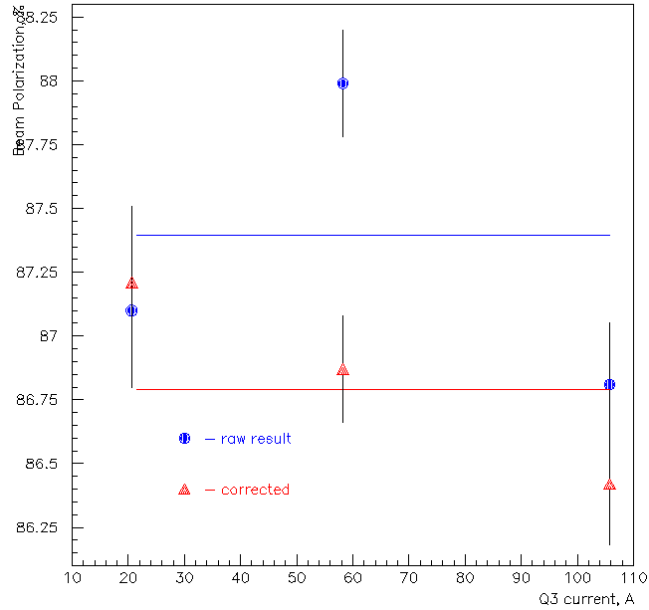
Analyzing power at Ebeam=11GeV



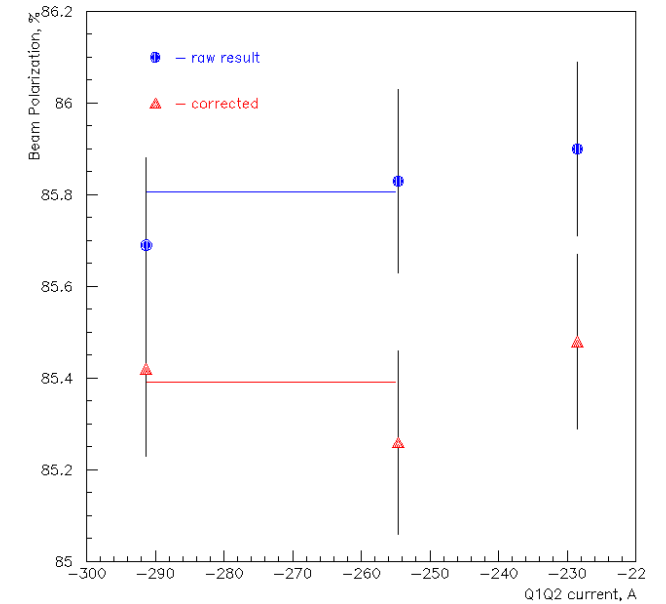
At 1 GeV GEANT precision is more important vs. 11 GeV

GEANT Corrections

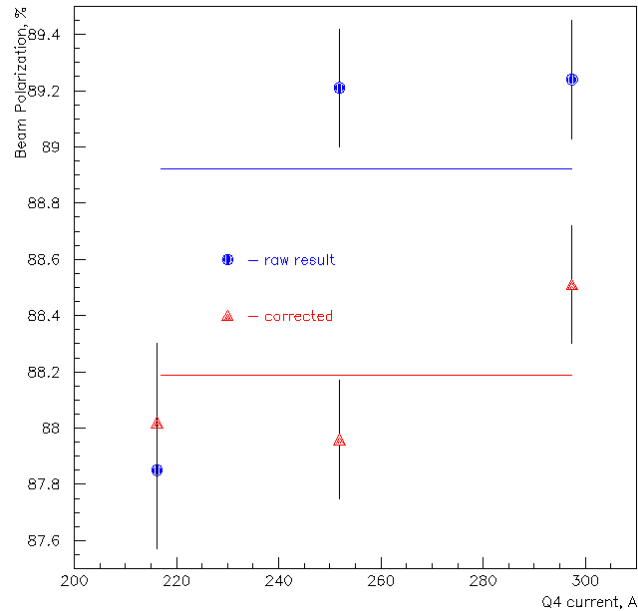
Moller results, 03/31/2016



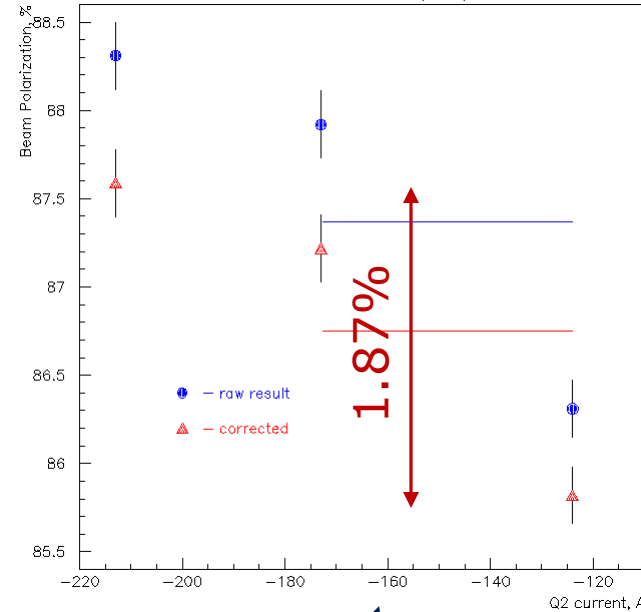
Moller results, 11/28/2016



Moller results, 04/09/2016

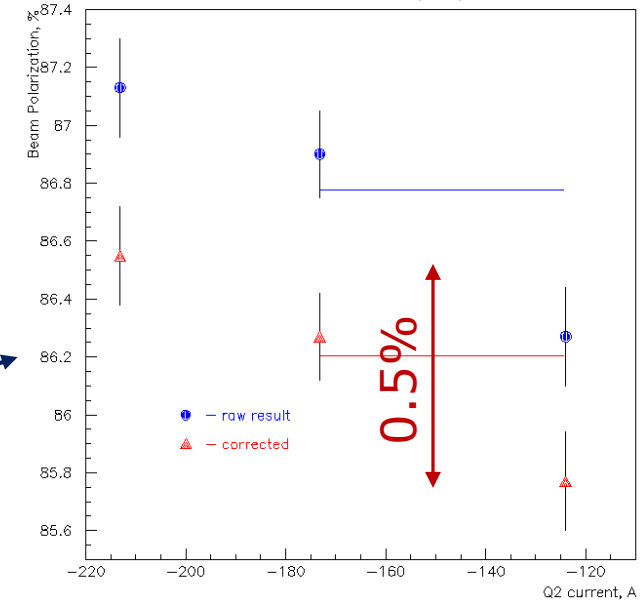


Moller results, 10/31/2016



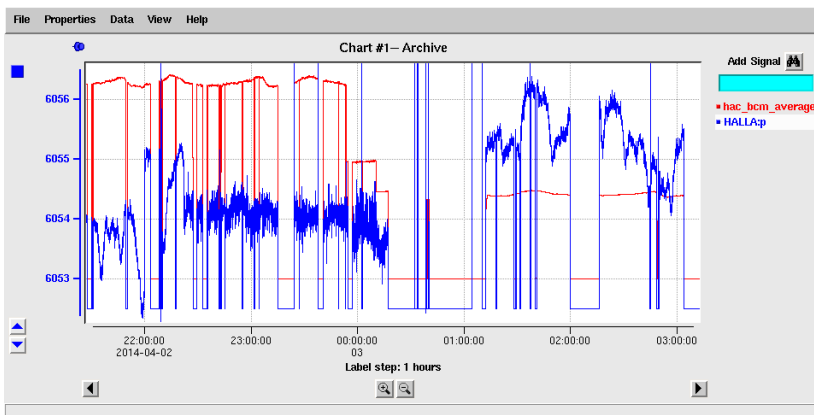
The same Moller settings at 8.5GeV

Moller results, 12/19/2016

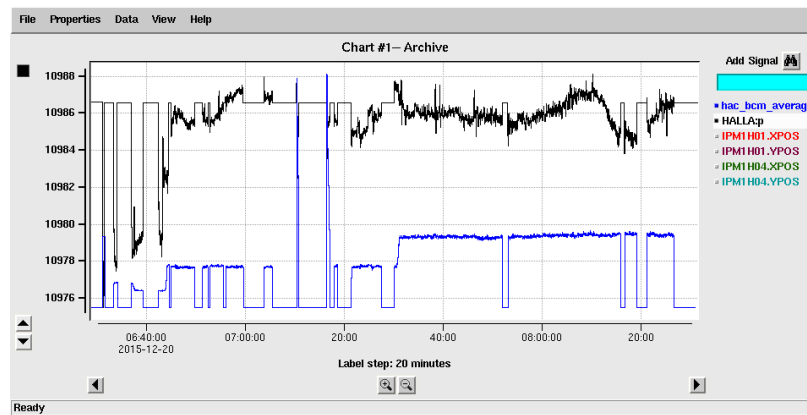


Beam Energy/Position Stability

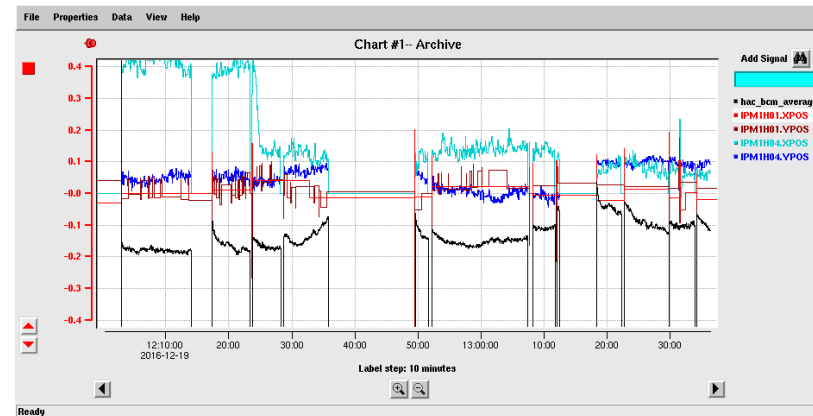
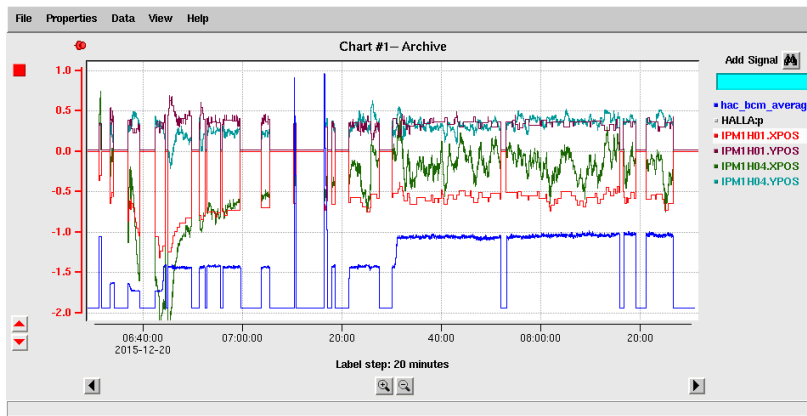
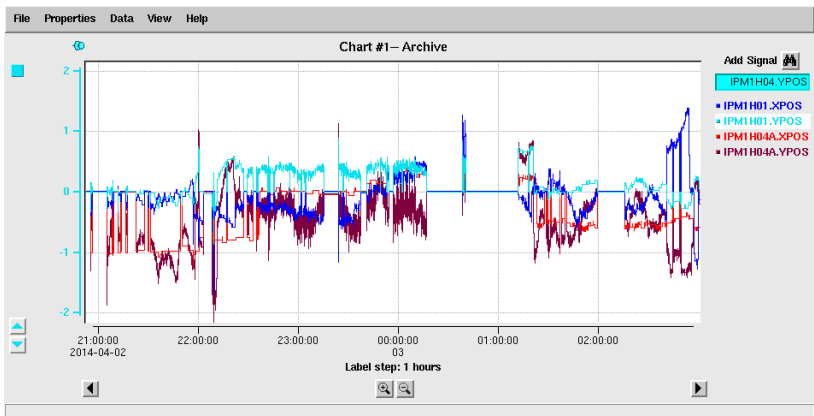
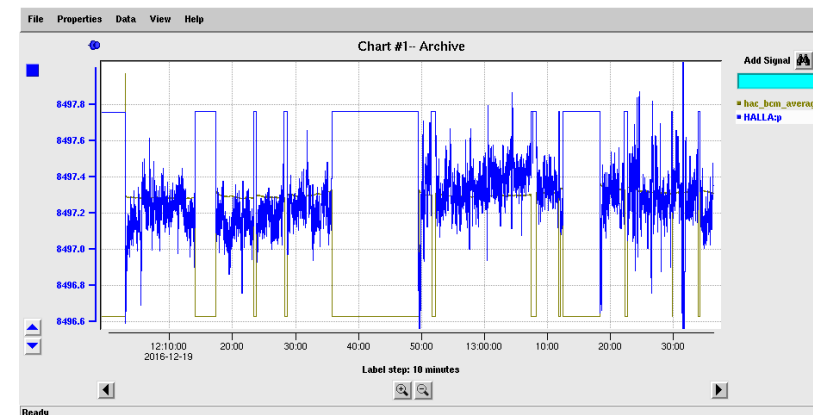
2014



2015



2016



Required beam position is $\pm 0.1\text{mm}$ at BPM 1H01 and 1H04

Highly required for successful precise Moller polarimetry:

- Cross check (spin dance) with Mott, Hall A and Hall C Compton and Hall C Moller
- Beat frequency test after injector upgrade to 4 halls running
- Target foils saturation measurement in Temple
- Kerr-effect apparatus

SUMMARY

2018 PREX/CREX:

- Don't have experience with the Moller polarimeter operation after upgrade at low beam energy (1- 2GeV)
- New target has to be commissioned
- New FADC DAQ has to be commissioned
- New superconducting magnet alignment test has to be done
- Low beam energy is bad kinematic for Moller polarimeter

Today the Moller polarimeter is not ready for PREX/CREX!
A lot of hard work have to be done!

SUMMARY - 2

Moller polarimeter before PREX-I:

- No any experience with a new superconducting magnet
- No any experience with a new brute force target
- No any experience with a new FADC DAQ
- No any experience with the Moller polarimeter at 0.8GeV

We are in better shape now!
We have a whole year before PREX/CREX!